

Mock Exam — Weeks 1 to 15

DSA27 · Data Structures and Algorithms · final practice · 60 marks: Section A 30 ·
Section B 30

Question bank practice paper

Fall 2026

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How to use this paper. Sit it in one go, closed book, with no computer, in the time the final gives you. It covers the whole course and leans on weeks 8–15, since the **weeks 1–3** and **weeks 1–7** mocks already cover the first half. The questions are taken from the weekly question banks, and **none** of them repeats either earlier mock, so you can sit all three. The model answers are in the weekly answers files — look them up **only after** you have finished and marked yourself.

The format follows the final: **30 marks of multiple choice, 30 marks of written answers.** The date and room are announced in the lecture and on the WhatsApp channel.

Section A — Multiple choice (30 marks)

One mark each. Choose one answer out of four. Write the letters on your answer sheet.

#	Question	#	Question	#	Question
1	W1-M03	11	W9-M10	21	W12-M17
2	W2-M04	12	W9-M17	22	W13-M02
3	W3-M05	13	W10-M02	23	W13-M10
4	W4-M05	14	W10-M10	24	W13-M17
5	W5-M06	15	W10-M17	25	W14-M02
6	W6-M06	16	W11-M02	26	W14-M10
7	W7-M07	17	W11-M10	27	W14-M17
8	W8-M04	18	W11-M17	28	W15-M02
9	W8-M12	19	W12-M02	29	W15-M10
10	W9-M02	20	W12-M10	30	W15-M17

The questions are in the weekly files, week01-questions.md to week15-questions.md — in the students' book, at the end of each week's chapter.

Section B — Written answers (30 marks)

Answer **all** questions. Marks are shown in brackets.

#	Question	Topic	Marks
B1	W8-E3	lower_bound and upper_bound: definitions and uses	4
B2	W9-T3	insertion sort traced, the array after every pass	2
B3	W10-E1	merge sort as divide and conquer; its recurrence	5
B4	W11-S1	a binary search tree drawn after ten insertions	3
B5	W12-T1	a min-heap after every push and pop	3
B6	W13-E4	tombstones in open addressing	4
B7	W14-T1	breadth-first search traced, queue and visited	3
B8	W15-E2	precedence is the shape of the grammar	4
B9	W10-B1 and W13-B3	two bugs, 1 mark each	2
Total			30

Coverage. Section A has seven questions from weeks 1–7 and 23 from weeks 8–15; Section B is entirely weeks 8–15, because the two earlier mocks give you written practice on weeks 1–7. The real final is not bound by that split: revise the first half from those two papers.

Marking yourself

- **Section A:** one mark per correct letter. The answer tables are at the top of each weekly answers file.
- **Section B:** use the **marking points** in the answers files. Award a mark for each point your answer makes clearly and correctly — not for length. For the traces and drawings (B2, B4, B5, B7), divide the marks evenly over the steps and award them for the steps you got exactly right.
- **Then** look at *which* questions you lost marks on, and at which level ([what], [how] or [why]). That tells you what to revise, and how — and the weekly question bank of that week has more of the same kind.
- **Remember the rule that fails students:** you need $\geq 30\%$ of the final as well as $\geq 60\%$ overall. Coursework cannot rescue a final below 30%.